

AALBORG AP, Denmark

WMO: 060300

Lat: 57.096N

Lon: 9.851E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	20.2	11.9	10.1	19.4	15.5	12.1	24.1	35.8	43.8	32.2	42.9	4.5	20	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	78.9	65.3	75.6	63.9	72.7	62.6	67.4	74.7	65.7	72.6	64.1	70.4	10.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.5	70.3	63.1	86.3	68.7	61.4	81.4	67.2	31.8	74.6	30.5	72.5	29.3	70.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	24.6	22.0	DB	9.8	83.4	5.7	2.7	5.7	85.3	2.4	86.9	-0.8	88.4	-5.0	90.3	(4)
(5)			WB	9.2	70.4	5.5	1.7	5.2	71.6	2.0	72.6	-1.1	73.6	-5.1	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	33.8	34.2	37.9	45.4	52.4	58.2	63.0	62.7	56.5	48.8	41.5	36.1
	DBStd	11.98	6.87	6.64	6.19	5.71	5.68	4.76	4.75	4.36	4.52	5.38	5.89	7.58
	HDD50	2299	503	443	377	160	37	1	0	0	4	84	258	432
	HDD65	6418	968	863	842	588	392	212	98	98	255	501	704	897
	CDD50	1431	0	0	1	22	112	247	403	394	200	48	4	0
	CDD65	74	0	0	0	0	2	8	36	27	1	0	0	0
	CDH74	461	0	0	0	0	16	72	241	128	4	0	0	0
	CDH80	53	0	0	0	0	0	9	36	8	0	0	0	0
Wind	WSAvg	10.7	11.8	11.9	11.6	10.4	10.0	10.5	9.5	9.7	10.3	10.6	10.7	10.8
Precipitation	PrecAvg	24.5	1.8	1.2	1.4	1.4	1.8	2.3	2.5	2.7	2.6	2.6	2.4	2.0
	PrecMax	34.4	4.0	3.5	3.4	3.2	4.7	4.9	7.9	8.0	7.0	6.0	4.7	4.3
	PrecMin	17.0	0.2	0.1	0.0	0.2	0.2	0.5	0.2	0.4	0.8	0.6	0.6	0.4
	PrecStd	4.2	0.9	0.7	0.7	0.7	0.9	1.3	1.6	1.5	1.2	1.3	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	49.0	56.0	67.3	76.7	80.9	84.2	81.0	73.7	64.0	56.7	51.8
		MCWB	46.6	46.0	48.2	56.1	62.7	64.5	66.2	65.5	64.0	58.4	54.2	50.1
	2%	DB	46.3	46.4	51.4	63.0	71.3	75.5	79.9	77.3	69.7	60.4	53.7	49.1
		MCWB	44.5	44.5	45.9	52.5	59.4	62.6	66.0	65.0	62.0	56.8	51.8	47.6
	5%	DB	44.6	44.6	48.4	58.8	66.7	71.3	76.6	74.7	66.5	58.3	51.7	47.4
		MCWB	42.9	42.8	44.3	51.0	57.2	60.7	64.8	64.1	60.3	55.4	49.8	45.9
	10%	DB	42.9	42.9	46.1	55.1	63.0	67.4	72.9	71.3	64.1	56.4	49.8	45.7
		MCWB	41.4	41.4	43.2	48.7	55.2	59.2	63.0	63.1	58.9	53.8	47.9	44.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	46.8	49.7	57.6	64.4	67.5	69.7	69.4	65.7	60.0	54.7	50.2
		MCDB	48.1	48.0	53.9	64.8	73.2	75.7	78.7	75.9	70.4	62.6	56.1	51.3
	2%	WB	44.8	44.7	46.9	54.0	61.1	64.5	67.7	67.2	63.3	57.5	52.2	47.9
		MCDB	45.9	45.9	50.0	60.6	68.5	72.5	76.1	73.4	68.3	59.7	53.6	48.8
	5%	WB	43.0	43.0	45.0	51.4	58.3	62.2	66.0	65.5	61.4	55.8	49.9	46.1
		MCDB	44.1	44.1	47.5	57.1	65.1	69.1	73.5	71.6	65.3	57.7	51.3	47.1
	10%	WB	41.5	41.5	43.4	49.4	56.0	60.1	64.3	63.9	59.7	54.3	48.1	44.3
		MCDB	42.7	42.7	45.6	54.2	61.8	65.8	70.9	69.5	63.2	56.3	49.4	45.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.0	10.6	13.9	14.7	13.8	14.3	13.7	12.1	9.9	8.0	7.6
		MCDBR	7.6	8.3	13.3	19.7	21.1	21.0	20.6	18.8	15.6	10.8	7.9	7.5
		MCWBR	7.4	7.3	9.1	11.4	11.7	11.0	9.6	9.1	9.7	8.0	7.3	7.3
	5% WB	MCDBR	7.4	7.7	11.9	17.7	18.1	17.8	17.7	15.6	13.9	9.8	7.6	7.4
		MCWBR	7.3	7.1	8.8	11.0	11.1	10.6	9.6	8.9	9.3	8.0	7.2	7.4
Clear-Sky Solar Irradiance	taub	0.308	0.333	0.351	0.388	0.380	0.371	0.390	0.392	0.365	0.347	0.324	0.294	
	taud	2.224	2.285	2.322	2.274	2.351	2.400	2.367	2.371	2.431	2.400	2.289	2.198	
	Ebn at Noon	175	224	254	261	272	276	267	258	248	218	168	145	
	Edh at Noon	17	24	30	36	36	35	35	33	27	21	16	13	
All-Sky Solar Radiation	RadAvg	123	299	749	1277	1701	1868	1718	1344	906	435	161	87	
	RadStd	25	53	67	146	153	142	166	94	90	55	29	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.62	N/A	N/A	N/A	N/A	N/A	-360	-596	+220	N/A
Regional (46 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	-214	-281	N/A	N/A

Nomenclature: See separate page